Tasmiah Haque

Ph.D. Candidate · Industrial Engineering · West Virginia University

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Research Interests

Generative time-series modeling for real-time AI systems under deployment constraints, with a focus on inference-time refinement and applications in bandwidth-efficient video motion synthesis, remote patient monitoring, and smart manufacturing.

Education

West Virginia University, Morgantown, WV
Ph.D. in Industrial Engineering — Aug 2023 – Nov 2026 *(Expected)*
Dissertation: *Generative Sequence Forecasting for Real-time Decision Systems Under Deployment Constraints*

West Virginia University, Morgantown, WV
M.S. in Industrial Engineering — Aug 2021 – May 2023
Thesis: *A Hybrid Physics-Infused LSTM for Marine Vessel Tracking via Automatic Identification System Data*

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
B.Sc. in Industrial and Production Engineering — Feb 2015 – Apr 2019

Publications

Refereed Journal Articles

- **Haque, T.**, Syed, M.A.B., Jeong, B., Bai, X., Mohan, S., Paul, S., Ahmed, I., and Das, S. (2026). Towards efficient real-time video motion transfer via generative time series modeling. *Multimedia Tools and Applications*, 85(2), p. 51. [Link]
- **Haque, T.**, Syed, M.A.B., Das, S., and Ahmed, I. (2024). Advancing marine surveillance: A hybrid approach of physics-infused neural network for enhanced vessel tracking using automatic identification system data. *Journal of Marine Science and Engineering*, 12(11), p. 1913. [Link]

Refereed Conference Papers

- **Haque, T.** and Das, S. (2025). Inference-time stochastic refinement of GRU-Normalizing Flow for real-time video motion transfer. *59th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, pp. 1313–1318. [DOI]
- Bai, X., **Haque, T.**, Mohan, S., Cai, Y., Jeong, B., Halasz, A., and Das, S. (2024). Enhancing bandwidth efficiency for video motion transfer applications using deep learning based keypoint prediction. *International Conference on Engineering Applications of Neural Networks (EANN)*, Cham: Springer Nature Switzerland, pp. 134–151. [Link]

Preprints

- Rahaman, M.M., Chang, E., **Haque, T.**, and Das, S. (2025). Post-training quantization of generative and discriminative LSTM text classifiers: A study of calibration, class balance, and robustness. *arXiv preprint* arXiv:2507.09687. *(To be submitted.)* [arXiv]

Manuscripts in Preparation

- **Haque, T.**, Kosinski, J., Al-Mamun, M.A., and Das, S. Unsupervised keypoints for real-time fall detection: Comparative analysis under real-world conditions with predictive bandwidth reduction. *(Manuscript in preparation.)*

Research Experience

Graduate Research Assistant Aug 2023 – Present
School of Mathematical and Data Sciences, West Virginia University
Supervisor: Srinjoy Das, Ph.D.

- **Bandwidth-Efficient Video Motion Transfer via Keypoint Prediction** [*EANN 2024*]
  Designed a Variational Recurrent Neural Network (VRNN)-based keypoint prediction framework integrated with the First Order Motion Model (FOMM), enabling efficient video frame synthesis with up to 2× additional bandwidth reduction. Applicable to video conferencing, virtual-reality gaming, remote medical monitoring, and anomaly detection in manufacturing.
- **Inference-Time Stochastic Refinement of GRU-Normalizing Flow** [*Asilomar 2025*]
  Developed a training-free MCMC-based stochastic refinement strategy for GRU-Normalizing Flow (GRU-NF) models, improving the diversity–fidelity tradeoff in generative video synthesis without retraining; integrated into a real-time, bandwidth-efficient motion transfer pipeline.
- **Generative Time-Series Modeling for Real-Time Video Synthesis** [*MTAP 2026*]
  Conducted a systematic accuracy–diversity tradeoff analysis using VRNN and GRU-NF models (PyTorch, PyTorchTS) for keypoint-based video conferencing, remote patient monitoring, and manufacturing anomaly detection.
- **Real-Time Remote Fall Prediction via Motion Keypoints** [Draft in Preparation]
  Extending the unsupervised keypoint motion framework to extract and forecast patient activity patterns in real time for fall detection, anomaly classification, and communication-overhead reduction in remote health monitoring. Mentoring an undergraduate co-author through the full research cycle.

M.S. Student and Researcher Aug 2021 – May 2023
Dept. of Industrial and Management Systems Engineering, West Virginia University
Supervisor: Imtiaz Ahmed, Ph.D.

- **Physics-Infused LSTM for Marine Vessel Tracking** [*JMSE 2024*; Presented at *IISE Annual Conference 2023*]
  Developed a hybrid algorithm combining physics-based motion models with LSTM for AIS-based vessel track association using TensorFlow, achieving 95% average prediction accuracy.

Teaching & Mentoring

Research Mentoring

Undergraduate Research Mentor May 2025 – Present
School of Mathematical and Data Sciences, West Virginia University

- Mentoring an undergraduate student, B.S. in Computer Science & Engineering, on the design and evaluation of an unsupervised keypoint-based fall detection system with real-time predictive bandwidth reduction.
- Guiding the student through literature review, experimental design, implementation in PyTorch, and academic writing; the student is a named co-author on the resulting manuscript currently in preparation.

Teaching Assistantship

Graduate Teaching Assistant
Advanced Analysis of Engineering Data (undergraduate course) Aug 2021 – May 2023
Dept. of Industrial and Management Systems Engineering, West Virginia University

- Designed and delivered weekly lab sessions in R covering exploratory data analysis, regression modeling, hypothesis testing, and statistical process control.

Topics in Deep Generative Models & Scalable Computing (graduate course) Jan 2026 – May 2026
School of Mathematical and Data Sciences, West Virginia University

- Created PyTorch Jupyter notebooks to support course instruction on linear neural networks for regression and classification, recurrent neural networks, and attention mechanisms.

Industry Experience

Graduate Engineering Trainee – Inbound Logistics Dec 2019 – Mar 2021
Unilever Bangladesh Ltd., Kalurghat Factory, Chittagong, Bangladesh

- Managed inventory safety and quality for 400+ SKUs; monitored stock levels in SAP to align with production scheduling.
- Optimized warehouse storage layout and transportation costs through cross-functional stakeholder coordination.
- Recovered $6,000 in packaging materials through a targeted relabeling initiative, meeting On-Time-In-Full delivery targets.

Technical Skills

Languages	Python, R, Matlab, C
ML / DL	PyTorch, PyTorchTS, TensorFlow
HPC	GPU-accelerated training & inference on NVIDIA RTX, A10, A100
OR / Sim	MPL/CPLEX, Arena
Engineering	AutoCAD, SolidWorks, SAP
Productivity	LaTeX, MS Office (Word, Excel, PowerPoint, Visio)
OS	Linux, Windows

Honors & Awards

Six Sigma Green Belt Certificate, *Institute of Industrial and Systems Engineering* Jan 2023

Supply Chain Director's Award, *Unilever Bangladesh Ltd.* Jun 2020

Relevant Coursework

Application of Neural Networks
Design of Experiments
Additive Manufacturing
Statistical Machine Learning
Quality and Reliability Engineering
Applied Linear Programming

List of References

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Dr. Imtiaz Ahmed
Assistant Professor, Department of Industrial and Management Systems Engineering
West Virginia University
email: imtiaz.ahmed@mail.wvu.edu